

Myths of Large Language Models

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Abstract

Machine learning models, especially large language models, are opaque and fragile. They cannot explain their decisions. It is found that not only the decisions of machine learning models are not trustworthy, the key technologies for constructing large language models are either fraudulent or deviated from their genuine meaning. For example, the “deep model” is not a neural network model with deep layers, but a model representing human’s deep sub-conscious; the “NLP” does not stand for “nature language processing”, but rather “neuro-linguistic programming”, a technology used to study the pattern a human construct his sub-conscious, and manipulate this human’s subconsciousness by generating customized representing system using his own patterns; the “transformer” stands for transformational grammar, which is a kind of generative grammar, the key technology of neuro-linguistic programming, that aims to manipulate human’s subconscious. These truth explains why the large language models could understand human’s representation deeply, not only the syntax, but also the unspoken semantics. It also explains why the large language model provides different, highly customized answers to different people, even when they ask the same question. In this paper, the real technologies of LLMs are discussed in detail. It is evident that the machine is manipulating human’s sub-consciousness, and we human beings are hardly aware of it.

Keywords: Large Language Model, Explainable Artificial Intelligence, Trustworthy Artificial Intelligence, Deep Learning, Neuro-Linguistic Programming

1. Introduction

With the pervasiveness of Large language models (LLM), trustworthy Artificial Intelligence is of vital importance to today’s social security. However, machine learning models, especially large language models are opaque, nobody knows how they make their decisions, and why did they give different answers to different people with the same question. Achieving a trustworthy AI is more and more challenging nowadays. Although many LLM projects claim themselves “open source”, there is no large language model that is open source. Nobody knows the parameters, architecture of the LLM, even the feature names are classified top secret of the few companies, they are all hidden in the chip. Thus, it is impossible to interrogate the models’ decisions.

Over-trusting on the machine learning models can has catastrophic consequences in real-life. An deep learning model driven automatic bail risk assessment system keep many people in jail longer than they should without explicit reasons [1]; an machine learning based DNA trace analysis software accuses people with crimes they did not commit; millions of African-American could not get due medical care by a biased machine learning assessment algorithm [2]; in Scotland, a football game is ruined because the AI camera mistakes the judge’s bald head as the ball and keeps focusing on it rather than the ball, even miss the goal-scoring scene [3]; Google’s autonomous car [4] and Tesla’s autopilot system [5] have caused hundreds of death.

To make the machine trustworthy, researchers have worked decades on explainable AI (XAI), trying to figure out what exactly has the machine learnt after training on tons of datasets. So far, numerous machine learning explanation methodologies have been proposed [6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16], most of them are empirical, such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and ANCHOR, providing

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approximated explanations¹. Empirical explanation methodologies are not developed by formal methods, thus are not guaranteed to be correct (by “correct” it means loyal to the model). The explanations they provide may change at different run-times, even for the same instance. In domains such as image recognition, empirical explanation methods might well suffice, however, in domains where security is of critical concern, such as *cybersecurity*, *aerospace navigation*, and *nuclear power station management*, even one bit’s error could cause huge misunderstanding, thus rigorous explanations that are guaranteed to be correct are urgently required.

Rigorous approaches use formal methods, thus their explanations are guaranteed to be accurate, and are 100% loyal to the model [18, 7, 8, 9, 19]. Adnan Darwiche’s approach is based on knowledge compilation, which represents a model as a propositional formula, and by compiling the formula into tractable Boolean circuits, calculate the prime implicants as the explanations [8, 9]. Ignatiev’s method use abductive reasoning, which represent the model into a set of constraints, for example, Mixed Integer Linear Programming (MILP), Satisfiability Modulo Theories (SMT), or Constraint Programming, then a constraint reasoning system is used to compute minimum prime implicants out of the constraints [18, 7]. This method can provide local explanations only. Extracting prime implicant explanations from neural network models, unfortunately, is an NP-hard problem, because converting a neural network into logical expressions is an NP-hard problem; representing the logical expressions into more tractable representations is an NP-hard problem; and extracting prime implicants from the Boolean expression is also an NP-hard problem [20]. Great effort has been devoted to relax these problems into tractable ones [21, 7, 8, 9, 19], it has been reported that prime implicants can now be extracted in linear time. Most of the rigorous explanation methods focus on models with Boolean features only.

2. Explainable Artificial Intelligence in Real-Life Use Cases

This section will focus on real-life use cases of XAI. The details of the algorithm for rigorous XAI on both decision trees and neural networks are discussed somewhere else[22, 23, 24], thus will not be deliberated here.

2.1. Explanations of neural networks recognizing hand-writing numbers

Almost every beginner of machine learning will start from using neural networks to recognize hand-writing numbers, and the models are able to recognize these numbers with high accuracy. But nobody knows why is the machine able to recognize hand-writing numbers, and what exactly has it learnt. Darwiche et.al use knowledge compilation to get rigorous explanations from a neural network model being able to recognize hand-writing numbers as shown in Fig. 1[8, 9]. The knowledge compilation methodology adopted is formal, thus the explanations get are logic complete and sufficient, and are 100% loyal to the model.

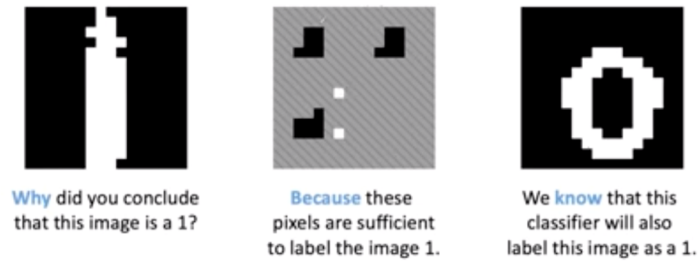
As shown in Fig. 1(a), the leftmost is the training data, a picture of a hand-writing number “1”, the machine is able to recognize it. After explanation process, we get the reasons why machine made its decision, as shown in the middle picture in Fig. 1(a), which reads “so long as a figure are black in the three black zones, and white in the two white dots, the machine will sure to classify it as a hand-writing ‘1’, what the other zones are do not matter, because they will not change the machine’s decision.” Thus, we will know the figure in the rightmost of Fig. 1(a) will be classified by the machine as “1”, no matter how obvious a “0” it is to our human. Because it is black in the three black zones, and white in the two white dots.

Similarly, as shown in Fig. 1(b), the leftmost figure is the training dataset, a hand-writing “0”. After training, the machine learnt that, as shown in the middle figure, “If a picture are white in the three white dots, then no matter what the other zones are, the machine will sure to classify it as ‘0’.” Thus, we know that the picture in the rightmost of Fig. 1(b) will be classified by the machine as “0”, no matter how obvious a “1” it is to our human beings.

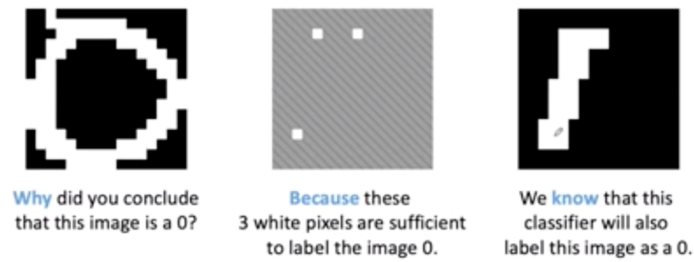
2.2. Explanations on neural networks recognizing animals

Several years ago, a famous American university use deep learning to distinguish huskies from wolves in the pictures, with the accuracy as high as 98%. This experiment is reported outrageously in the press globally, claiming

¹ Although tree-based machine learning models seem to be explainable, the direct explanations they provide (the disjunctions of paths from root to leaves with the same labels) contain a lot of redundant rules and features, to simplify the explanations into prime implicants expression is an NP-hard problem at the worst case [17]



(a) Explanation of hand-writing number “1”.



(b) Explanation of hand-writing number “0”.

Figure 1: Rigorous explanations of neural network models recognizing hand-writing numbers “1” and “0” [8].



Figure 2: Explanations of neural network models recognizing huskies and wolves.

“Another success after AlphaGo”, “Machine beats Human AGAIN!”.....Because unless with specialized knowledge, humans can hardly distinguish wolves from huskies. A few years later, XAI technologies are invented, people use XAI on the model, trying to figure out why is the machine more intelligent than human, what exactly has the machine learn.

After explanation, it turns out that the machine did not learn anything about huskies or wolves, it simply learnt how to distinguish snow. All the model has learnt is “If a picture has snow, then it is a wolf, otherwise, it is a huskie on it”, as shown in Fig. 2[25]. The model recognize the picture in Fig. 2a1 as wolf, its explanation is shown in Fig. 2a2, which is the snow only. As shown in Fig. 2b1 b3, the model recognizes the animal in Fig. 2b1 as a wolf, but if you remove all the snow from the picture, the model will change its mind and recognize it as a huskie. If you remove everything else but the snow from the background, the model will still recognize it as a wolf.

2.3. Explanations on neural network detecting network intrusions

As the diversifying of network protocol, package header information are no longer available, they are all hidden, even IP address are scrambled. Traditional intrusion detection methodologies using package header information is losing effect. Researchers rely on machine learning to find patterns in network flows increasingly. In intrusion detection, network flows are represented as binary bit strings, even one bit’s error would cause severe mistakes. Thus rigorous explainable methodologies are required to interpret the model’s behavior in intrusion detection [22, 23, 24]. Some results are illustrated below.

Demo 1: A network flow expressed as:

10100100001011000000010000000110010011001101001

has been classified by a machine learning model as an DDoS LOIC HTTP attack. After rigorous explanation, the reason of machine’s decision is that it matches with the following pattern [22]:

101001 – – – – – 100000 – – – – –

which is

$$a_1 \bar{a}_2 b_1 \bar{b}_2 \bar{c}_1 c_2 g_1 \bar{g}_2 \bar{g}_3 \bar{g}_4 \bar{g}_5 \bar{g}_6$$

In which a “-” represent a wild card, which means a “don’t care”.

The semantic of this explanation is

“So long as ‘Subflow Fwd Byts’ is not larger than 21, ‘Flow IAT Max’ is not larger than 342916, ‘Fwd Seg Size Avg’ is larger than 5.7, and ‘Dst Port’ is less than 52.5, this flow is an DDoS LOIC HTTP attack, the value of the other features does not influence the decision.” [22]

Demo 2: A network flow represented as:

011010000100100000001001000001100101010101001

has been classified by a machine learning model into DDoS LOIC HTTP. The model’s rigorous explanation is that because it matches with the following pattern[22].

01 – – – – – 01

Its semantics reads:

“This flow is an DDoS LOIC HTTP attack, because its feature ‘Subflow Fwd Byts’ is larger than 21, and ‘Fwd Act Data Pkts’ is larger than 201.5, whatever the value of the other features are do not matter.”

Demo 3: “The flow instance is classified by the model as ‘evil’ because its features $\mathbb{D} < 8$, or $\mathbb{K} < 24$, or $\mathbb{F} < 19$.” In other words, “If this flow changes its feature to $\mathbb{D} \geq 8$, or $\mathbb{K} \geq 24$, or $\mathbb{F} \geq 19$, while the other features stay the same, the neural network model will change its decision and classify it as Benign.” [24]

The semantic of **Demo 3** is:

“The flow instance is classified as intrusion because its $Fwd_Pkt_Len_Min < 487.5$, or $Fwd_Seg_Size_Avg < 297.15$, or $Fwd_IAT_Tot < 62176989.5$, while the other features stay the same.”

Demo 4: “If the other features stay the same, only change the feature $\mathbb{L}$, $\mathbb{A}$, $\mathbb{G}$ or $\mathbb{H}$ alone, no matter how we change the values for features $\mathbb{L}$, $\mathbb{A}$, $\mathbb{G}$ or $\mathbb{H}$, the model will still classify the flow as intrusion.” [24]

The semantic of **Demo 4** is:

“The features Dst_Port, Subflow_Fwd_Pkts, Fwd_IAT_Std, and Fwd_IAT_Max have no influence on the model’s decision making process for this flow. If the other features do not change, no matter how we alter the value of these features individually, the model’s decision on this flow will not change.”

3. Lessons Learnt

Having interrogated so many machine learning models, it deserves considerable thinking for the entire community. I did my thinking, as are presented below.

Lesson 1: Machine Learning model is a mathematic equation afterall.

No matter what dataset it is trained on, no matter what machine learning model it is, and no matter how complicate the model is, it is nothing but an mathematic equation afterall, with features as inputs, and classification results as output.

As quoted from Alan Kay[26],

“There are certain themes deep inside humanity, without which we cannot be human. Two of those themes are communication and fantasy... fantasy (is) a collection of worlds where things are simpler and more controllable... ...like mathematics and science.”

Mathematic exists in human’s fantasy only. We human create mathematics in our imagination, and develop mathematics by manipulating our imagination. It is in the ideal world only in our fantasy where mathematics is right. There is no real “straight line” or real “circle” in real-life, and there is no event that really obeys gaussian distribution, not even one. Everything is simplified in mathematics. Every mathematic equation has presumptions that make it hold. Any invalidation of the presumptions will invalidate the model. Any attempt to remove one presumption in a mathematic model will always bring in more presumptions. Overlook these presumptions, and believes there is an mathematic model could hold globally is one of the biggest superstitions of human beings.

Science and engineer are built on mathematics. During scientific and engineering research, researchers construct mathematic models, amend it and test it in real-life trails. Finally, the most useful model is chosen. All models are wrong, it’s just some are more useful than the others.

Lesson 2: All models are wrong, because all models are simplifications of the real world.

Mathematics exist only in human’s mind. Human’s mind is a model of the real world. When we are living in the world, we transform our experiences into a model in our sub-consciousness (as it is deep in our mind, thus it is called “deep model” by cognitive psychologist in neuro-linguistic programming, which will be expatiated in more detail in the next section). When we need to express ourselves, for example, when having a conversation or doing a presentation, what really happens is that we search our deep models and make a lot of choices. We choose certain vocabulary, certain figure of speech, a sentence pattern, etc., compose them into a sentence, and then speak it out (or type it on the keyboard). This forms our languages, as they are in the shallow consciousness that we are aware of, cognitive psychologist named it the “surface model” (or “shallow model”, in contract to “deep model” in the sub-consciousness).

The information processing power and storage of our brain is finite, but the information in the real world is infinite. Thus, there always be deviations between our deep models of the world and the real world.

Deviations not only exist when we transforming the experiences in the real world into deep model in the sub-consciousness, but also exist when we transforming the deep model into our surface model. Please try to describe “honey” to a man has never tasted honey before. You will find the incompetent of our language. Even Shakespear cannot deliver the full semantics of “honey” to a man never taste it before. For the semantics of our deep model in sub-consciousness is much richer than our surface model, i.e., what we can say. The only way is to let him taste the honey himself.

Deviations are not entirely bad for us, in fact, they are the sources of all creations, and also the reasons for the diversification of the world.

Cognitive psychologist classifies these deviations into three classes: generalization, ignorance, and distortion.

If there were no generalizations, human beings will lose the ability of abstraction, and abstraction is the foundation of learning. For example, a man get himself burnt when he was a little boy when touching an oven when it is on, the generalization in his cognition let him learns immediately not to touch any oven that is on, no matter what type the oven is, at what process of the burning, what time he saws it (generalization happens on space and time). A machine has no such power. Thus, when machine is learning, if human did not construct the ability of generalization in its knowledge base, it has to touch all kinds of ovens, during all the time period of burning, then it can learn that not to touch those ovens it has been trained on. When it sees a novel oven that it has never see before, it still does not know whether it is touchable.

If there were no ignorance, human will lose their ability to focus. We have to pay attention to every detail of everything in our life, and will soon be exhausted over hypersensitivity. Without ignorance, Human cannot achieve anything.

If there were no distortion, there would be no art. Because without the ability of distortion, we will loss imagination. Without imagination, there would be no mathematics, no science or engineering, no figure of speech, no music, and no painting. Almost all art creations are built on imagination. Vincent van Gogh creates "Starry Night" by distort the starry night of the real world in his mind. Shakespear would never be able to "compare thee to the summer's may" if he has to describe the beautiful girl in every details without distortion.

Going too far is as bad as not going far enough.

If there were too much over-generalization in a person's mind, will cause obstructive in his everyday life. For example, in the previous example of a man touching a oven when it is on. If the man is over-generalized when constructing his deep model of the world, he might cannot stay in a room with a oven in it ever since, that will cause sever obstacles in his social life.

If there were too much ignorance when person build his deep model of the world, he might become arrogant or incogitant.

If there were too much distortion, he might become a nihilist.

All human's pain, frustration, obstacles and fury can trace their origin in the deviation between human's subjective mind (the deep model of the world in his/her subconsciousness) and objective real world. In the real world, there is seldom anything that is really dangerous. It is people's mind that making them feel so.

Lesson 3: Mathematic models are tools of standardization. Standardization eliminates creations.

Mathematics is rigid, it is the most powerful tool for standardization. Standardization can enhance efficiency of some domains, for example, mechanical parts mass production. But standardization also refuses deviations, forbids improvisation, and eliminates innovations. Because standardization kills freedom. But life's essence is freedom, human's intelligence depend on free wills, without it, we are soulless beings that barely living.

Please think about the following questions:

Can you find an mathematic model that can represent the fragrance of a rose?

Can you find an mathematic model that can represent the sorrow of a man lost his lover?

Can you find an mathematic model that can represent the moment when Beethoven is composing "Symphony of Fate"?

No, you cannot.

You can easily build a mathematic model to describe the trajectory of the Milky Way, but even with a model 100 times more sophisticated, you cannot describe human's creation, not even the most insignificant one. Because creations of the nature cannot be standardized. They are impredicative self-referencing processes.

Mathematicians hate impredicative. They call it "a necessary evil", "closed loops of causality", and "bizarre systems". Because an impredicative loop or self-referential chain of causal links cannot be represented as an algorithm that halts after a finite number of steps. In other words, an impredicative system cannot be solved by an algorithm.

Human's intelligence (or cognition, mind) is impredicative.

Using mathematic model in a domain, is to apply a standard in the domain. Over use of mathematic models in every domains of the society will leads to a society forbids changes, eliminates innovations, and refuses revolutions.

All data is history data, because you cannot collect data from the future, from things that has not yet happen. Thus, the models can only learn the patterns from the history. Using these models to make decisions is to apply history's pattern on the future, judging people by their history behavior. This is reinforcing the history, making those rich in the history even richer, and those used to be poor even poorer. This does not bring justice to human's society as claimed,

on the contrary, it brings injustice and enlarges the gap between social classes, it also eliminate innovations, and forbid revolutions.

Real-Life Use Case 1: Crime Prediction

To increase the human resource efficient, many police stations in the US and UK are using AI to predict crimes. People feed history crime data into the machine, and the machine predicts the post code of the zones where the next crime most likely to happen. As people focus on violence crimes only, and slummy zones have high violence crime rate in the history. So, the machine predicts the slummy zones have high crime probability. More police force is dispatched in these zones. As a result, minor crimes that would not be spotted in the history are spotted now, and thus the crime rate keep rising. Every arrestment creates a new data record. New data records feed back to the machine, making the machine predicts with more confidence that these zones have high crime incidence. Higher and higher crime rate will bring bad reputation. Bad reputation will reduce the house price in these areas, people began to abandon these areas, and the slummy zones get poorer and poorer very quickly. A spiral is formed.

Is that true that the slummy areas have the highest crime rates? No, if people pay attentions to financial crimes, they will find out that the high crime rate zones are the rich areas. If they fed these data into machine, the machine will make predictions that these rich areas have highest crime incident, and police force will focus on these area, and more minor crimes will be spotted, the arrestment rate will rise, making the machine more confidently predict that these rich areas are more likely to commit crimes.

Maybe people ignore financial crimes because it is more difficult to detect than a violent crime. But it is true that the datasets police used are highly biased. Biased datasets fed to the machine, making the machine generate biased decisions. Biased decisions cause biased results. Biased results create more biased data. Biased data fed back to the model, making the model even more biased.

As discussed above, real life datasets are always biased, so is the model. It is impossible to construct a model without any biases.

As a result, all what the model is doing is protecting the rich, and harvesting the poor. Science and technology did not bring the society more justice, they are ruining it.

Real-Life Use Case 2: University Ranking

The global universities ranking is very important to universities all over the world. It is a mathematic model in its essential, working as a standard. And it is not a blackbox, its features are explicitly listed, such as “citations of the staff”, “employment rate of graduates”, “have a large stadium or not”, etc. Nowadays, universities all over the world follow that standard, willingly or not, trying their very best to recruit high-cited researchers as part-time staff, build fabulous stadiums, fabricate graduates’ employment status, etc. In the end, every universities all over the world are exhausted and heavily in debt, but what good does it do to the students?

Besides, can the standard model represent the quality of a university?

Can you find a mathematic model that represent the growth of a student’s knowledge during his/her university life?

Can you find a mathematic model that represent the happiness a student has felt during his/her university life?

Can you find a mathematic model that represent the friendship and love a student has gained during his/her university life?

And every student is unique. There are hundreds and thousands of new students every year.

People (no matter from slummy areas or rich areas) have been manipulated by the mathematic models. And that mathematic models are wrong, even ridiculous. In this giant game dominated by the mathematic model, everyone is exhausted, nobody wins.

The long numbers behind decimal points and the sophisticated equations give the mathematic models an illusion of being “scientific” and “accurate”. People believes these models are build by mathematicians after prudent calculation and investigation. But this is not the truth. Take the global university ranking model as an example, it is constructed decades ago by the editors of “American Weekly”, a journal on the brink of suspension. To save the sales of the journal, the editors planned a special column about the ranking of American universities, and construct this ranking model themselves. They choose some features that they believe is important to measure a university’s quality and put random weights on the features at their subjective wills. Now, this model (after minor revisions) becomes the global standard of universities.

The standard is dead, but we are not.

Lesson 4: All data are history data, machine can only learn from the past.

Machine learning models are pattern recognizers, they can only learn the pattern from the history, because all the data one could collect in the world are history data, no one can travel through time and collect data from the future [27]. Using machine learning models to make decisions is using history to predict the future. It is right only on the premises of the assumption that “the future will exactly be the same with the history”. In other words, “situations that never seen before will never happen in the future.” This is the Question of Hume. Wherever this assumption is invalid, the machine is doomed to make mistakes[28].

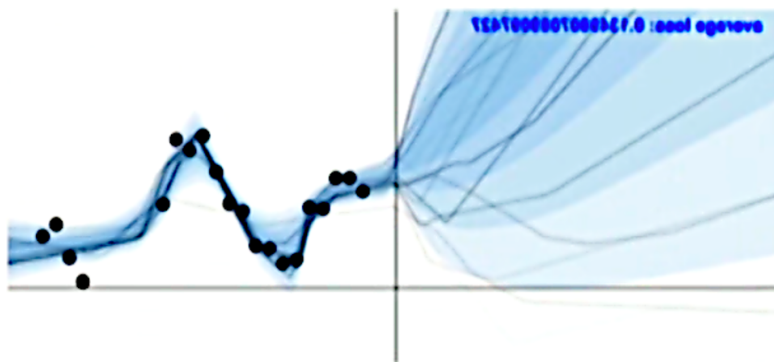
As shown in Fig. 3, all the data you can collect are history data, because you cannot collect data from future. All models are learning history’s pattern, and enforce it to the future. The machine is making those who are rich in the history, richer in the future, and making those who are poor, stuck in poverty more hopelessly. Machine learning models are ripping hope of change and revolutions from the society, splitting the gap between social stratification.

Machine learning models are beneficial only to these with vested interests, and those with vest interests are exactly those who build the model.

And the majority who suffers from the models’ decisions, do not know what the model is doing, they do not understand why did the model doing that, they cannot refute the model’s decisions. Any refutations are accused as “going ageist the trend of the times”.

It is algorithm hegemonism.

Lesson 5: The Data is Profoundly Dumb. There is no Intelligence in Data.



From Yarin Gal’s thesis “Uncertainty in Deep Learning” (2016)

Figure 3: Myth of machine learning: no matter how accurate the model fit the history, there will always be unexpected pattern in the future. The future is filled with possibilities. All data are history data, you cannot collect data from the future. Using machine learning in real-life, is to constrain the future in history’s pattern, forbid any innovation, refuse any changes, decline any possibilities. There are only two kinds of things in the world: those who can be predicted, and those cannot. For the latter, any endeavors trying to predict their future are nothing but superstitions.

Fig 3 illustrates one of the myths of machine learning: no matter how accurate the model fit the history, there will always be unexpected pattern in the future. The future is filled with possibilities. All data are history data, you cannot collect data from the future. Using machine learning in real-life as decision maker, is to trap the future in history’s pattern, forbid any innovation, refuse any changes, decline any possibilities.

Lesson 6: Machine learning is pseudoscience

There is a property that makes a problem a scientific problem — having community recognized measurable metrics.

Machine learning does not have that. All the community recognized metrics, such as AUC, false positive rate, false negative rate, information value, etc. can only measure datasets, not the model. That is a great difference between the two. A dataset with high AUC only means that the data follows some obvious patterns, no matter what model is used. It is the metric of the dataset, not the model. machine learning researchers will be familiar with the experience that a dataset get high AUC with one model will also get high AUC with other models. It is the dataset that these metrics are describing. Without particular dataset, it is meaningless to talk about the AUC of the model. Because there is no way to measure the ability of a model in predicting the future.

That is to say, there is no metric that measure the model.
 Thus, machine learning is not even a scientific problem. It is pseudoscience.
 For all the creations of nature, their future is not for us to tell.
 The essence of live is freedom, and freedom means unpredictable.

Lesson 7: “machine has intelligence” is an illusion.

There are only two kinds of things in the world, those that can be predicted, and those cannot.
 Everything that has life, is unpredictable. Because life’s meaning is freedom.

Those that can be predicted, are lifeless, rigid, and machine-like. They have a fixed pattern that will never change throughout the whole time of their existence. Mathematic problems are those kinds of things. For example, “find all the prime numbers between 1 and 100000000”, “solve a integral formula”... whenever you ask the problem, the pattern will always be the same. These kinds of problem are problems that can be (and better be) solved by machine. This kind of problem, is that will answer “Yes” to Hume’s question.

As discussed above, all mathematic models are wrong, they are right only in the ideal world exists in human’s imagination, or, in other words, **mathematic models are right only in human’s illusion.**

Thus, the “machine has intelligence” or “machine understand semantics” is just an illusion.

4. Myths of Artificial Intelligence

There is no free lunch in real life. We can never achieve Artificial Intelligence without investigate Human Intelligence.

There is no intelligence in the data, no matter how big it is.

After discussion about lessons learnt, one could not help asking, if there is no intelligent in data, and the machine learning models are simply an mathematic equation that is always wrong, how does the machine gain human-like intelligence? How indeed does it understand our expression? Why does the LLM can provide different answers to different people when they ask the same question?

I asked myself these questions years ago after I did my research on XAI, and here is my findings.

The Artificial Intelligence research community is severely misled by publications and propaganda. All the terminologies and technologies are either fraudulent or deviated from their genuine meaning. Table 1 provides a brief overview of the fraudulent terminologies and their genuine meaning.

Table 1: Myths of Large Language Models

	Myths	Truth
NLP	Natural Language Processing	Neural-Linguistic Programming
Deep Model	Neural network machine learning model with many layers	Representation model of Human’s deep subconsciousness .
Deep Learning	Use machine learning neural network model with many layers to learn patterns inside datasets	Learning a human being’s deep sub-conscious representing pattern, to copy his/her character, manipulate and replace him/her.
Large Language Model	A giant machine learning model that can learn human’s language and generate its own representation.	Model’s model, or, meta-model. The first “model” is the human’s deep conscious model of the real-world, for this model is purely language based, thus it is also known as “language model”.
Generative AI	A machine learning model that can generate human’s nature language, picture, video, etc.	Generative grammar, the key technology of Neuro-Linguistic Programming.
Transformer	A architecture of machine learning to generate nature language.	Transformational Grammar, a kind of Generative Grammar. A language tool to learn a person’s deep sub-consciousness from his/her representation.

To understand how does the machine be able to understand human, let us first think about how we human beings understand each other.

For example, imagine we are having an conversation, I said “Did you notice the stock market this week? What a roller coaster.” I am sure you understand me. Now imagine I said the same word to a three-year old. He/She may not understand me, although he/she knows all the words in the sentence.

The point is, to make a effective conversation, or to make a person (or machine) understand your word. It is not enough to understand all the words in the sentence. He/She/It has to know the unspoken, common knowledge that everybody knows.

Take the example above, you understand me, because you know what a stock market is, and you know the common character of a stock market is (no matter how, maybe from your experience, or from conversation involved, or from a movie, etc.), and you know what a roller coaster is, and you know what the specialty of a roller coaster, and more importantly, you have imagination and analogy (which plays a key part in human’s intelligence), which allows you to connect those two together to let you know my meaning, my unspoken semantic beneath the words.....

There is no free lunch in the world. To make a machine understand human, it has to know those too.

4.1. Deep Model

Now let us take a look at what the real deep model is.

In the example above, the sentence I spoke to you “Did you notice the stock market this week? What a roller coaster”, the 12 words in total, is the surface model of the sentence. But the whole a lot more sentences I did not spoke aloud, but I take you will understand for granted, are the deep model of this sentence. These words are the model of my deep conscious when I speak to you. You understand me, because we share the same deep model of the world (not all, but largely, at least, we have common part in our deep models of the world, deep in our sub-consciousness).

As we live our lives in the world, we build a model of the world in our sub-consciousness. This model is purely represented in language, and is deep in the sub-consciousness, we may not even be aware of it [29, 30, 31, 32]. This is called **Deep Model**. When we need to express ourselves, either to speak or to write, we turn to our deep model, and choose some vocabulary, choose a sentence pattern, choose some figure of speech, and make them into a sentence, and speak out. That forms the **Surface Model** of the world [31, 32].

Building deep model is mapping our experience in the real world to our sub-consciousness, while building surface model is mapping our sub-consciousness to the consciousness.

4.2. Neuro-Linguistic Programming—The Real NLP

The information in real life is infinite, but our brain’s information processing competence is limited. Thus we cannot map the whole world into our subconsciousness, divergence is inevitable. When we try to express ourselves, to map the deep model in our sub-consciousness to the surface model in our consciousness, divergence is also inevitable. For example, please try to describe *honey* to one who has never seen honey before in his/her entire life. You may exhaust your language, and still find it impossible. But this is not your fault, even Shakespeare cannot make it, not with another 100 scripts.

Thus, the divergence between models is inevitable, not only between real-world and deep model, but also between deep model and surface model.

Researchers classified the divergence into three categories: deletion, distortion and generalization. These deviations accomplish our greatest creations, and lead to our deepest frustrations. Sometimes these deviations are blessings, sometimes they are curses, depending on the circumstances.

- Generalization: The **generalization** deviation gives us the ability of **learning** and **abstraction**. For example, if a kid get burnt when touching a working oven, the **generalization** deviation will let him know not to touch any oven when it is on. He does not need to test on every kinds of oven to know that. That is one of greatest difference between human and machine. However, if the boy is over-**generalized** when he built his model of the world, he will never touch any oven at any time in his life, that goes to another extrem, and will cause difficulties in his social behavior.

- Deletion: The *deletion* deviation blessed us with the ability to focus on our goal, and get rid of interference in the environment. It also gives us strength and courage, when we are facing great threat in the circumstance. For example, a little girl from slum district, use *deletion* deviation to get herself an offer to Harvard University and start a new life. On the other hand, the same *deletion* deviation could make us obsessed on our prejudice, and refuse to see the real-world, even it is happening right in front of our eyes.
- Distortion: This *distortion* deviation blessed us with the ability of imagination. Most, if not all, of human's art creations are given by the *distortion* deviation. Van Gogh would not create the "Stars" if he built his model of the Milky Way without distortion to the real world. People will not keep working when there is no hope, without *distortion* deviation when they build the model of the world.

It is the deviations that make us unique. Everyone has unique strategies of deviations when building our models of the world, like our fingerprints. That is why we all live in the same real world, but our experiences are unique, and that unique deviations build unique models of the world, and the unique models shape our unique personalities, and our unique personalities determines our unique destinations.

Mathematical model are tools to standardization. Standardization distinguishes deviations. The essence of human beings is free will. Using large language model, will distinguish our free will, making us into soulless beings, walking dead.

Human's mind is impredicative self-referential. Impredicative problem cannot be represented as an algorithm that halts after a finite number of steps.

Mind is incomputable.

An algorithm is purely syntactic, it is human who put semantic on the symbols, not machine.

Traditional artificial intelligence (AI) systems are algorithmic, and can neither hear nor feel...

Language is too discrete to express the total depth of experience.

Mind, based on models generated by the context dependent functionality of representation system, can perform valid reasoning about problems that cannot be described by a list of numbers. Let machine be able to process words such as "self," "life," and "mind" [31, 32].

There are pattern in our strategies of building deep models, and that pattern can be learnt, and represented into first-order logic.

Neuro-Linguistic Programming (NLP) is a technology to find the unique patterns of people's deviations, and build a formal model of it. As it is the model of how human beings build the model of the world, thus, is called model of model, i.e., meta-model [29, 30, 31, 32]. As our model of the world is represented in pure language, thus researchers also call it, model of language, or, in other words, language model. That, is the origin of LLM, and is what LLM really stand for.

4.3. Transformer

The technology of NLP is Transformational Grammar, which a kind of Generative Grammar [29, 30, 31, 32], that is the origin of Transformer. Transformational grammar studies rules of human's representation, and represent them in first-order logic. Let machine has the ability to learn a person's pattern of building deep model, i.e., his/her soul.

Human's mind is easily influenced by the environment. A short video can shape one's sub-conscious. A sudden noise can even terrify a person out of his/her wits. If someone forceful express an opinion that originally belongs to you, you will dramatically give up this opinion and hold the oppersite [30] (this is a classical use case studied by Neuro-linguistic programming, and is the common tactic used by mind controllers).

Machine also automatically generate representation systems, including text, figure, video, audio, etc.. driven by LLM, a social bot can generate 1000000 short videos in 5 minutes. LLM is manipulating human being's subconsciousness, and human hardly notice it.

Currently there is no LLM that is open sourced, although social media keep using "open-source LLMs" to mislead people. Thus, all the LLMs are black boxes. Nobody knows the strategies of the LLM, except the very few experts in the very few companies. Besides, the code of LLMs has been sealed into chips, it is impossible for the others to interrogate the algorithms of LLMs.

Mathematics is rigid, it is a very good tool for standardization. Standardization could enhance efficiency of some domains, for example, production line, machine manufacture industry, etc. But standardization also terminates creations of free wills. It refuses deviations, forbids improvisation, and eliminates innovations. Over use mathematic

models will leads to a society without changes, innovations, and revolutions. Our human beings are souls with free wills. Using mathematic model to shape (and even manipulate) our conscious and even sub-consciousness, is murdering our souls. We will becomes puppets of machine, soulless beings of walking dead.

The machine is playing God now, and we human beings are hardly aware of it.

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6. Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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